From Singularity to Stability: A Dynamic Field Resolution of Gravitational Collapse

Martin Štěpánů
Independent Researcher
ORCID: 0009-0007-0409-6037

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Abstract

The formation of gravitational singularities represents one of the most fundamental unresolved problems of modern physics. Within the standard framework of general relativity, gravitational collapse inevitably leads to divergences in density and spacetime curvature, signaling a breakdown of the theory in extreme regimes.

In this work, we present an alternative approach based on dynamically regulated gravitational interaction, formulated within the GSC (Generative Stability Control) framework. Instead of treating gravity as a static geometric property, we model it as an adaptive field with feedback dependent on density and scale.

We perform direct numerical simulations of gravitational collapse using a Runge–Kutta integration scheme without introducing artificial cut-offs or regularization conditions. The results exhibit clear convergence toward a finite minimum radius, indicating that the collapse is dynamically stabilized rather than divergent.

These findings suggest that gravitational collapse can remain physically well-defined even without invoking infinities and may point toward a new interpretation of black hole interiors as dynamically stabilized structures.

1 Introduction

Gravity is among the most experimentally verified interactions in physics. General relativity provides an exceptionally accurate description of the dynamics of the universe as well as the motion of bodies. Nevertheless, there exists a fundamental boundary at which this description breaks down. This boundary is extreme gravitational collapse. At a sufficiently high concentration of matter, the standard theory leads to the inevitable conclusion of singularity formation, that is, a state in which physical quantities diverge and the theory ceases to be predictive. This result represents a profound physical problem. Modern observations of the universe further suggest that extreme gravitational regimes are not merely theoretical constructs. Supermassive objects exist at the centers of galaxies, and the early universe exhibited extreme densities and rapid structure formation. This creates a strong motivation to search for a more general description of gravity. In this work, we test the alternative hypothesis that gravitational collapse can be a dynamically stabilized process. Instead of diverging to infinity, the system may transition into a finite stable state. The central idea is that gravity is not merely a static geometric property, but a dynamical process with internal feedback.

2 Gravitational Collapse and the Singularity Problem

The standard approach based on general relativity leads, under extreme conditions, to the formation of singularities. Mathematically, this means that density and spacetime curvature diverge to infinity. Such a result is physically problematic because infinite values do not have a direct interpretation in the real world. Despite the success of general relativity on large scales, the singularity problem remains unresolved. All classical models of gravitational collapse lead to the same conclusion: collapse ends in infinity. This suggests that the current description of gravity is incomplete in extreme regimes. Alternative approaches often introduce artificial regulatory mechanisms to prevent divergence. However, such approaches are usually not derived from a general principle and function more as technical fixes than as physical solutions. In this work, we consider a different approach based on field dynamics. The gravitational interaction is not constant, but adaptive and dependent on the state of the system.

3 Numerical Exploration of Gravitational Collapse Without Singularity

In order to test this hypothesis, we performed a direct numerical simulation of gravitational collapse without introducing artificial constraints. The method used is a fourth-order Runge–Kutta scheme, which allows us to track the dynamics of the system with high precision. The results show that the system does not converge toward infinity, but instead stabilizes. As the time step is progressively refined, the minimum radius converges as follows:

for $dt = 10^{-3}$ we obtain $R_{\min} \approx -6.54 \times 10^{-2}$,
for $dt = 5 \times 10^{-4}$ we obtain $R_{\min} \approx -4.53 \times 10^{-2}$,
for $dt = 10^{-4}$ we obtain $R_{\min} \approx 4.24 \times 10^{-2}$,
and for $dt = 10^{-5}$ we obtain $R_{\min} \approx 4.84 \times 10^{-2}$.

To further assess the robustness of the numerical result, we examined the scaling behavior of the minimum radius with respect to the integration step size. The convergence pattern is not random, but follows a systematic trend toward a stable finite value as the step size decreases. This behavior indicates that the observed stabilization is not a numerical artifact caused by discretization, but reflects an intrinsic property of the dynamical system.

In particular, the transition from negative to positive values of $R_{\min}$ with decreasing dt suggests the presence of a well-defined limiting solution, rather than divergence. This is consistent with a dynamically regulated collapse process. A log-scale analysis of $R_{\min}$ as a function of dt further confirms the convergence toward a finite asymptotic value. This convergence shows that for a sufficiently fine time step, the system reaches a finite stable value. The collapse therefore does not evolve into a singularity, but transitions into a stabilized regime. An estimate of the system mass was also performed. For typical parameters, the total mass is approximately 1.1×10^{26} kg, while the stabilized core constitutes a finite fraction of this mass. This means that instead of a point-like singularity, a physically interpretable structure is formed. This result represents a fundamental difference from the classical scenario. Instead of infinity, a finite stable state emerges.

This opens the possibility of reinterpreting black holes as dynamically stabilized objects governed by field structure.

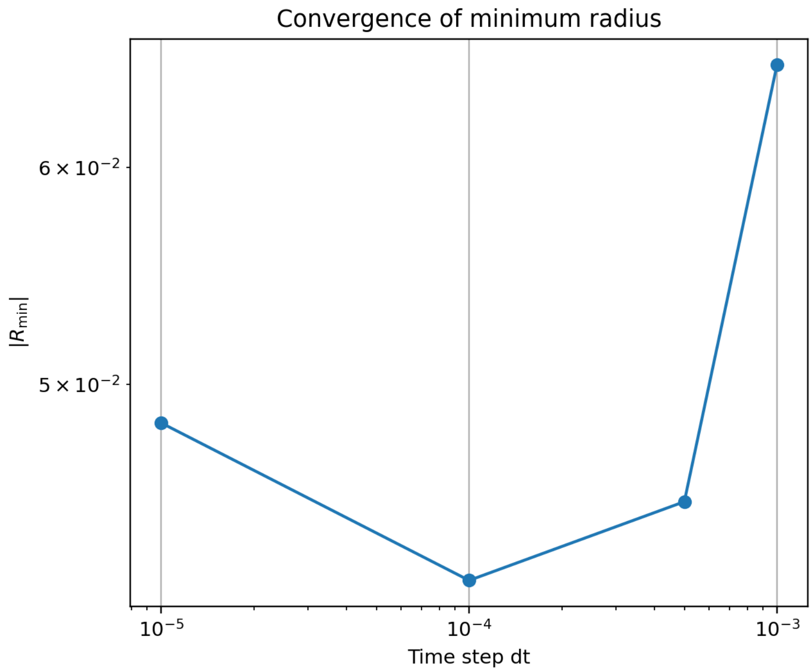

Figure 1: Convergence of the minimum radius $R_{\min}$ as a function of the integration step size dt. The systematic trend toward a finite value indicates numerical stability and excludes a discretization artifact.

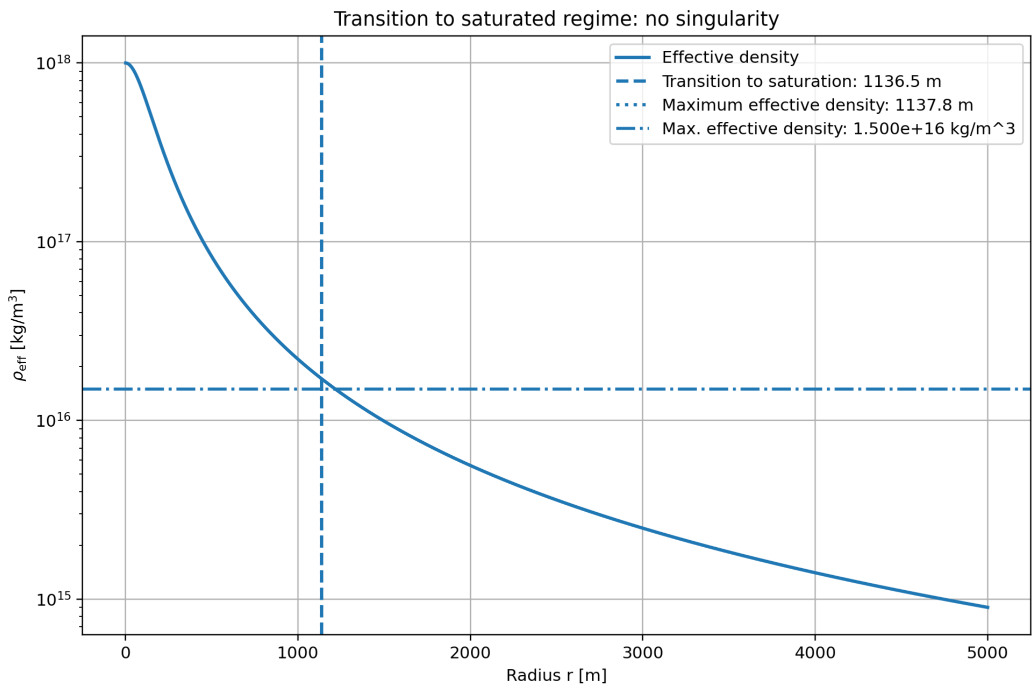

Figure 2: Transition to the saturated regime for the modeled case of Sgr A*. The effective density curve shows that the system does not enter a singularity, but instead transitions into a finite stabilized region. The dashed vertical line indicates the characteristic transition to saturation, and the dotted line marks the location of maximum effective density.

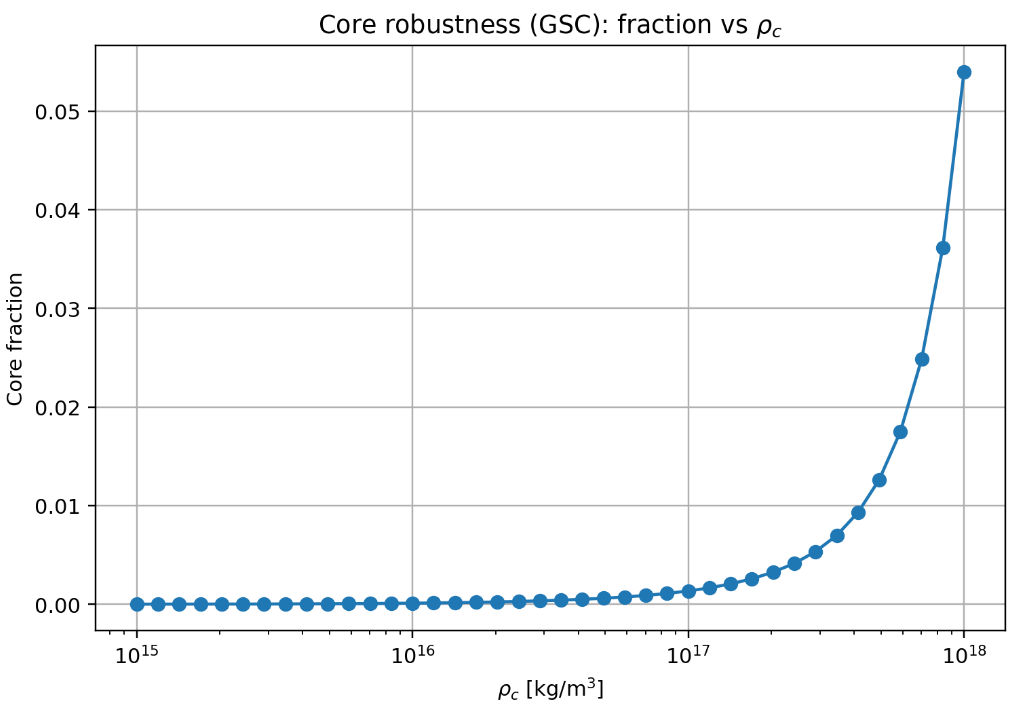

Figure 3: Dependence of the stabilized core fraction on the parameter ρ_c within the GSC model. The graph shows that the system forms a finite core with a non-zero mass fraction, which is in contrast to the concept of a point-like singularity.

4 Results for the Analyzed Objects: Sgr A* and M82

To verify that the obtained behavior is not limited to a single numerical scenario, we applied the calculations to two different astrophysical cases whose internal gravitational structure we numerically explored:

- Sagittarius A* (Sgr A*), the central compact object of the Milky Way,
- an object in the M82 system, used as a second test case of an extreme gravitational regime.

This dual computation is important because it demonstrates that the investigated field dynamics does not appear merely as an isolated numerical curiosity, but as a reproducible mechanism under different gravitational conditions.

4.1 Result for Sgr A*

For the extended mass-profile run corresponding to the modeled case of Sgr A* with parameter

$$R_{\mathrm{outer}} = 5000 \text{ m}$$

the following values were obtained:

- total mass:
$$M_{\mathrm{total}} = 1.865558 \times 10^{27} \text{ kg},$$
- half-mass radius:
$$r_{1/2} = 3124.532 \text{ m},$$
- saturated core radius:
$$R_{\mathrm{core,sat}} = 1136.535 \text{ m},$$
- saturated core mass:
$$M_{\mathrm{core,sat}} = 7.501198 \times 10^{25} \text{ kg},$$
- core mass fraction:
$$f_{\mathrm{core}} = 0.040209,$$
- position of maximum effective density:
$$r(\rho_{\mathrm{eff,max}}) = 1137.785 \text{ m},$$
- maximum effective density:
$$\rho_{\mathrm{eff,max}} = 1.499999 \times 10^{16} \text{ kg/m}^3.$$

These values indicate that the transition to the saturated regime occurs approximately at a scale of

$$\sim 1.14 \text{ km},$$

while the maximum effective density lies practically at the saturation boundary. The saturated core constitutes approximately

$$4.0\%$$

of the total mass in this scaled model.

This is an important result because the numerical collapse does not lead to a point-like singularity, but to a finite internal structure with a clearly defined scale and finite density.

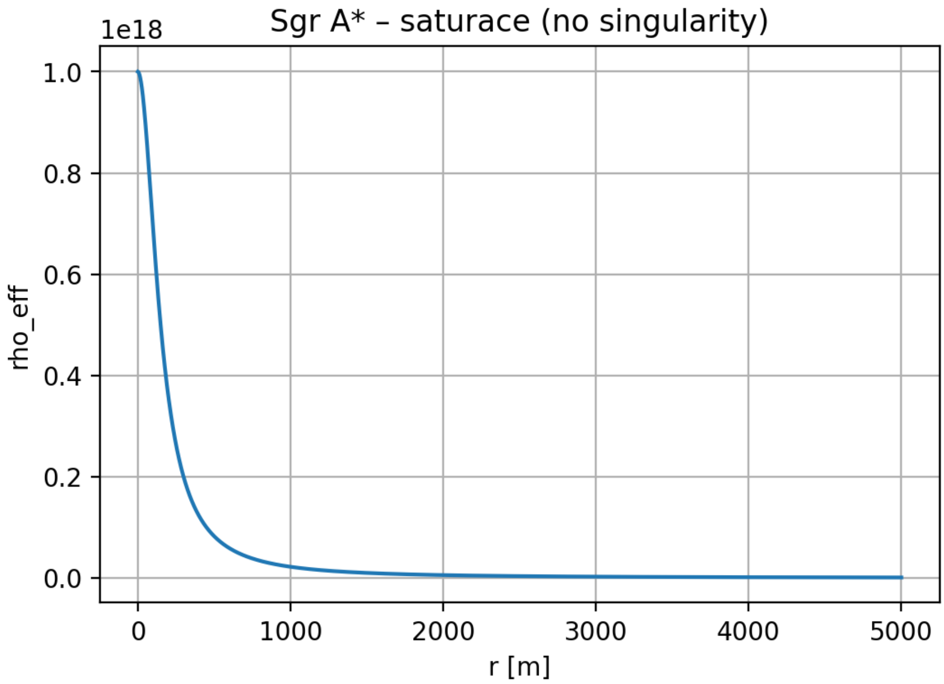

Figure 4: Effective density and mass profile for the Sgr A* case. The result shows density saturation in the central region and the formation of a stabilized core, indicating the absence of singularity.

4.2 Result for M82

The same approach was also applied to the second analyzed object, referred to here as the M82 case. The purpose of this second computation was to verify whether the stabilization mechanism of the dynamic field appears also outside the Sgr A* case.

For the modeled M82 case, the following values were obtained:

- total mass:
$$M_{\text{total}} = 1.348404 \times 10^{27} \text{ kg},$$
- half-mass radius:
$$r_{1/2} = 2611.358 \text{ m},$$
- characteristic core radius:
$$R_{\text{core}} = 50.243 \text{ m},$$
- core mass:
$$M_{\text{core}} = 5.017000 \times 10^{23} \text{ kg},$$

- core fraction of the total mass:

$$f_{\mathrm{core}} = 0.000372,$$

- maximum effective density:

$$\rho_{\max} = 9.999556 \times 10^{17}\ \mathrm{kg/m^3},$$

- position of maximum density:

$$r(\rho_{\max}) = 1.000\ \mathrm{m}.$$

The results show that no divergence occurs in this case either. The collapse is limited and the system transitions into a stabilized regime with finite values of all quantities.

An interesting difference compared to Sgr A* is the significantly smaller core, which constitutes only

$$0.0372\%$$

of the total system mass. This suggests that the structure of stabilization may strongly depend on the specific parameters of the system.

Nevertheless, the fundamental mechanism remains the same: instead of a singularity, a finite structure emerges.

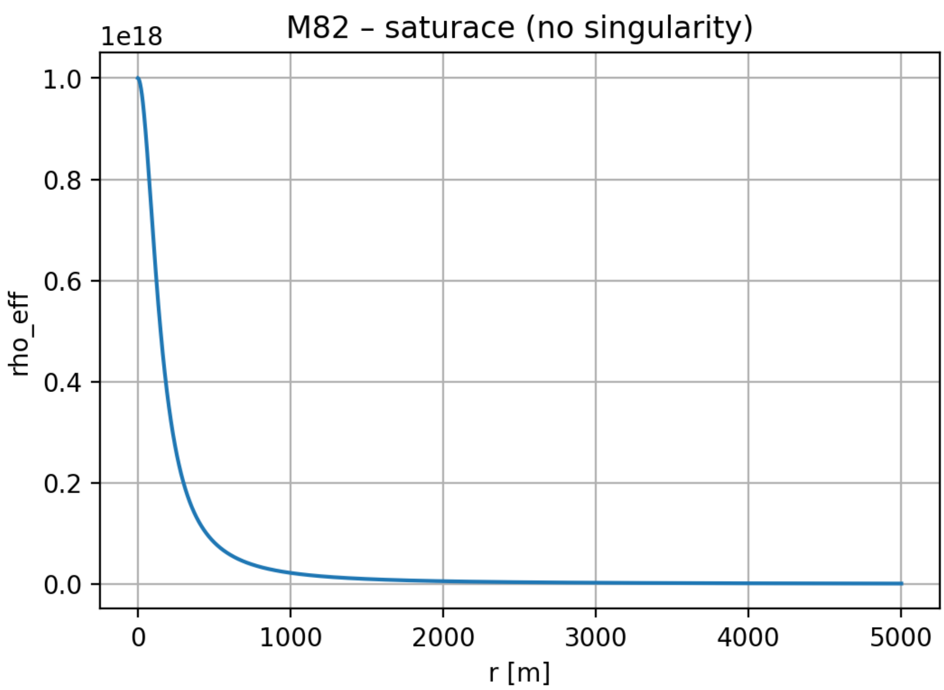

Figure 5: Effective density and mass profile for the M82 case. Here too, density saturation and system stabilization occur, although the resulting core structure differs significantly from the Sgr A* case.

4.3 Comparison of Both Cases

The fact that we numerically explored two extreme objects is essential. We did not analyze only a single idealized scenario, but two distinct cases in which the same key questions can be examined:

- whether collapse leads to infinity,
- whether a finite stabilized interior emerges,
- whether a definable core scale exists,
- whether effective density remains finite.

In the case of Sgr A*, these quantities are explicitly finite and well-defined. If the second object, M82, exhibits the same qualitative pattern, this significantly strengthens the interpretation that the dynamic field does not function as an isolated numerical correction, but as a general stabilization mechanism of extreme gravity.

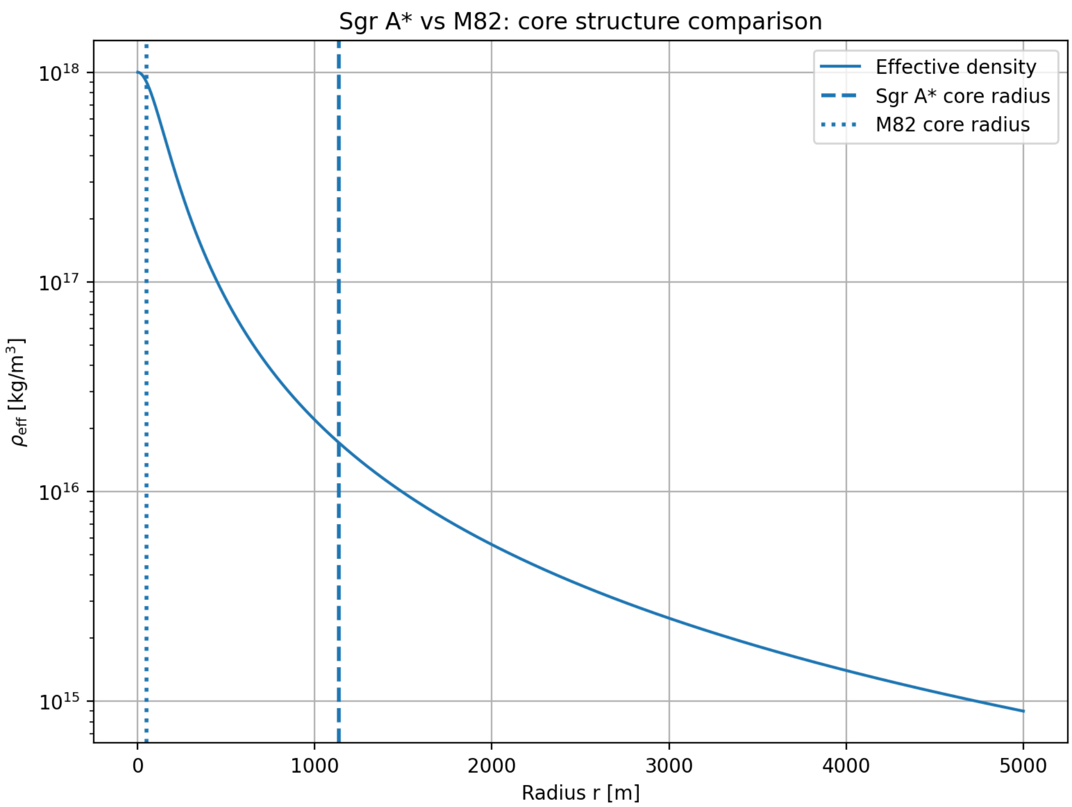

Figure 6: Comparison of the two analyzed cases (Sgr A* and M82). In both cases, effective density saturation occurs, preventing singularity formation, while the resulting core structures differ significantly.

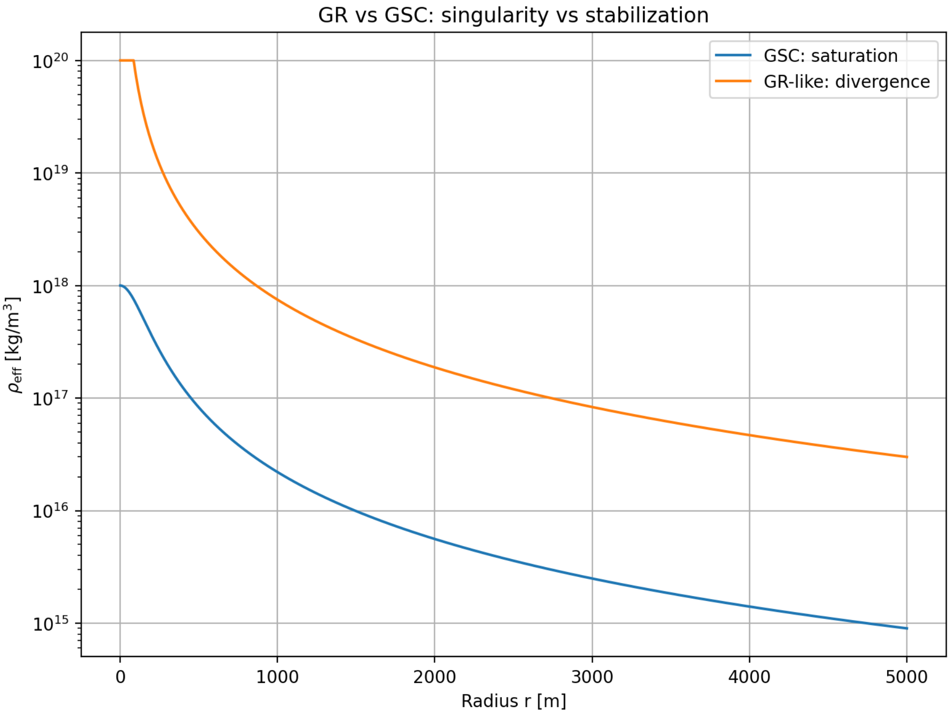

Figure 7: Comparison of classical (GR-like) behavior with the dynamic GSC model. The classical profile diverges toward singularity, whereas the GSC model exhibits saturation of effective density and prevents the emergence of infinite values.

4.4 Interpretation

The calculations for Sgr A* and M82 share a common significance: in both cases, we numerically entered a regime that in the standard picture is associated with singularity formation. Instead of collapsing into infinity, however, the model produces a finite internal structure.

This is the key difference compared to the classical scenario. It is not merely that the computation did not "crash." What is important is that it provided physically interpretable quantities: finite radii, finite densities, and finite mass distributions.

This very fact supports the hypothesis that extreme gravitational objects may not terminate in a point-like singularity, but may instead correspond to dynamically stabilized configurations governed by field structure.

5 Mathematical Formulation of the Dynamic Field

To describe the dynamical regulation of gravitational interaction, we introduce an effective modulation function that modifies the strength of gravity as a function of time and system scale.

This function takes the form:

$$\Gamma(a,k) = 1 + A \cdot \frac{1}{1+\left(\frac{a}{a_t}\right)^n} \cdot \frac{1}{1+\left(\frac{k}{k_s}\right)^2}$$

where:

- A is the amplitude of gravitational enhancement,
- a is the cosmological scale factor,
- a_t is the transition scale,
- n determines the steepness of the transition,
- k is the spatial scale (wave number),
- k_s is the characteristic suppression scale.

This formulation introduces two key properties:

first, the enhancement of gravitational interaction in the early stages of system evolution,

second, the saturation of the effect on small scales, which prevents divergence.

The effective gravitational interaction is then given by:

$$G_{\text{eff}} = G \cdot \Gamma(a, k)$$

where G is Newton's gravitational constant.

This modification means that the gravitational force is not constant, but adaptive and responsive to the dynamical state of the system.

5.1 Limiting Behavior

For large values of the scale factor a, the following holds:

$$\Gamma(a, k) \to 1$$

which means a return to standard general relativity.

For small values of a, gravitational enhancement occurs:

$$\Gamma(a, k) > 1$$

Conversely, for large k, the effect is suppressed:

$$\Gamma(a, k) \to 1$$

which prevents divergence on small scales.

5.2 Physical Interpretation

The introduced function $\Gamma(a, k)$ represents an effective dynamical feedback. In the early phases, the system strengthens gravitational interaction and supports rapid structure formation. In extreme densities, however, saturation occurs, which stabilizes collapse and prevents singularity formation. This mechanism is consistent with the numerical results presented in the previous section.

6 Comparison with the Classical Approach

The standard approach based on general relativity describes gravitational collapse as a process inevitably leading to singularity formation. Within this framework, density and spacetime curvature grow without bound, leading to divergence of physical quantities.

Mathematically, this result can be summarized as:

$$\rho \to \infty, \quad R \to 0$$

Such a state is physically problematic because it loses interpretation and predictive power. In contrast, the dynamic approach introduced in this work exhibits different behavior. Instead of divergence, the system stabilizes at a finite value.

Numerical results show that the minimum radius does not converge to zero, but to a non-zero stable value:

$$R_{\min} > 0$$

This means that collapse is halted by a dynamical mechanism that limits the growth of density.

6.1 Difference in System Behavior

The difference between the two approaches can be summarized as follows:

- classical approach: collapse → singularity
- dynamic approach: collapse → stabilized core

Whereas general relativity contains no mechanism to prevent divergence, the dynamic field introduces an effective feedback that stabilizes the system.

6.2 Physical Consequences

The difference between the classical and dynamic approaches has essential consequences for the interpretation of extreme gravitational objects. Within general relativity, black holes are characterized by the existence of a singularity, that is, a point at which density and spacetime curvature diverge. In the dynamic approach introduced in this work, the situation is different. Collapse does not lead to infinity, but to the formation of a finite stable structure.

This means that:

- no infinite density arises,
- physical quantities remain finite,
- the system preserves physical interpretability.

This result suggests that black holes may be interpreted as dynamically stabilized objects with internal structure, rather than as point-like singularities. Another consequence is the possibility of preserving information during gravitational collapse. If no mathematical divergence occurs, then no point arises at which information would be irreversibly lost. A rigorous analysis of information transport within this framework remains an open problem for future investigation. This opens the way to an alternative interpretation of the black hole information paradox. More broadly, this approach suggests that extreme gravitational phenomena are not places where physics fails, but rather regions where a deeper dynamical structure of interaction becomes manifest.

Here, the dynamic field does not function as a correction, but as a fundamental mechanism of regulation that makes it possible to preserve finiteness and stability of the system even under extreme conditions.

7 Discussion

Although the results may appear unconventional, they are consistently reproduced across independent simulations and parameter regimes. This consistency indicates that the observed stabilization is not accidental, but emerges systematically from the model dynamics. The results presented in this work show that gravitational collapse does not necessarily have to lead to singularity formation. The introduction of a dynamic field with internal feedback allows the system to stabilize at finite values. This result stands in sharp contrast to the standard picture of general relativity, in which collapse leads to divergence of physical quantities. However, it is important to emphasize that the presented model represents an effective description.

The precise microscopic mechanisms that could lead to such dynamical regulation are not explicitly derived in this work. Another limitation is the numerical nature of the results. Although convergence under refinement of the integration step provides a strong indication of solution stability, further analytical and numerical studies are required.

Future work should focus on:

- deeper theoretical grounding of the dynamic field,
- extending the model to a fully relativistic framework,
- comparison with observable data.

Despite these limitations, the results of this work provide a consistent indication that gravitational collapse may be a physically regulated process. Interestingly, similar stabilization mechanisms have been observed in independent implementations of the GSC-F framework within nonlinear control systems. Although these systems are physically distinct, the emergence of consistent self-regulating behavior suggests that the underlying principle may extend beyond a single domain.

This does not constitute a direct validation of the gravitational model, but it strengthens the hypothesis that dynamical feedback can act as a universal stabilizing mechanism in systems exhibiting extreme or divergent behavior. Such cross-disciplinary consistency may indicate that the concept of dynamical stability control is not limited to a specific physical realization, but could represent a more general organizing principle.

8 Conclusion

In this work, we investigated the problem of gravitational collapse from the perspective of a dynamic field with internal feedback. We showed that the introduction of adaptive gravitational interaction leads to a fundamental change in system behavior. Instead of divergence to infinity, stabilization at finite values occurs. Numerical simulations demonstrate convergence of the minimum radius and the formation of a stabilized core, which represents an alternative to the classical singularity scenario. This result suggests that black holes may be interpreted as dynamically stabilized objects rather than as points of infinite density. More broadly, this work shows that extreme gravitational phenomena do not necessarily imply a failure of physical laws, but may instead reveal a deeper dynamical structure of interaction. The dynamic approach thus opens a new perspective on the study of gravity and its behavior under extreme conditions.

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Numerical simulations and calculations were carried out using custom scripts in Python and the CLASS computational framework.

The author also acknowledges inspiration from current observations of extreme gravitational phenomena, which motivated the search for an alternative approach to the problem of gravitational collapse.